ARNAV VERMA

Dehradun, India | arnav04verma@gmail.com | Portfolio | Linkedin | GitHub | Medium | Youtube

PROFESSIONAL SUMMARY

Embedded Systems and **Unmanned Aerial Vehicle (UAV)** Engineer specialising in autonomous flight systems, avionics, and real-time control. Hands-on experience building and flight-testing autonomous UAVs using **PX4, ArduPilot, MAVSDK, MAVLink,** and **Raspberry Pi** companion computers, from **KiCad** PCB design and **embedded C/C++** firmware to Software-in-the-Loop **(SITL)** validated offboard autonomy. Strong foundation in path planning, sensor fusion, computer vision, and RF communications

Education

Thapar Institute of Engineering and Technology, BE in Electronics and Computer — Aug 2023 – Aug 2027

- CGPA: **9.34/10.0 |** Digital System Design, Analog Electronics, Computer Architecture, Embedded Systems, DSP

Work Experience

UAV Systems Intern, EME corps, Indian Army — March 2026 – March 2026

- Completed a 3 week internship with the **Indian Army's Electronics and Mechanical Engineering Corps** to work on a project **Hailing Swarm Drone System**
- Awarded a **Certificate of Internship** by EME for successful project delivery

Head of Avionics, Team Sammpaati — August 2025 – Present

- **Project Lead — NIDAR Disaster-Management Drone Competition (DFI)**; led a **10-member** multidisciplinary team for end-to-end development and integration of an autonomous UAV platform
- Conducted UAV workshops and trained **40+ juniors** in embedded systems and communication protocols

Projects

Fully Autonomous Heavy-Lift UAV for Disaster Response Flight Video

- Designed, manufactured, and flight-tested a custom coaxial UAV with a **1000mm wheelbase**, hybrid 3D-printed and carbon-fiber tube airframe, using **18" rotors** to achieve **22 min endurance** and **5 kg payload capacity**
- Integrated **Pixhawk** + **Raspberry Pi** companion computer; built KML-driven autonomous missions (area scan, survivor detection, payload delivery) and an onboard **OpenCV + ML** inference pipeline for real-time flood-survivor detection
- Enabled **4G/LTE** based long-range communication and autonomous payload release

Reefer Container Drone Inspection System

- End-to-end autonomous UAV inspection system: **BAPLIE D95B EDI** parser auto-generates a photorealistic **Gazebo** SDF simulation world from vessel stow-plan data with no hardcoded geometry
- **PX4 SITL** offboard controller via **MAVSDK** executes a 6-phase per-container flight pattern achieving below 0.35 m steady-state position error; Python **Tkinter** GUI provides live minimap, snake-pattern sequencing, and CSV audit logging

Offboard Mission Planning and Area Coverage Software for UAVs

- Developed a standalone offboard system converting **user-defined KML** regions into executable UAV missions via geometric scan-pattern generation and area decomposition for parallel **multi-UAV** coverage
- Implemented a MAVSDK based mission execution layer on a companion computer to upload, monitor, and manage long duration autonomous missions

Skills

Languages: C, C++, CUDA, Python, Pytorch, TensorFlow, TensorRT, ONNX
Embedded Systems: FPGA, STM32, ESP32, Arduino, UART, SPI, I2C, PWM
Companion Computers: Raspberry Pi 4/5, Linux (Ubuntu), Shell scripting, Jetson Nano, JetBot
Drone Autonomy and Control: PX4, Ardupilot, MAVLink/MAVSDK, DroneKit, SITL, Gazebo, Python
Communication and Networking: Wi-Fi,Bluetooth, Adalm Pluto SDR, HackRF One, GNU Radio, GQRX,
Tools and Platforms: GitHub, Docker, VS Code, MATLAB/Simulink, KiCad, MAVProxy, QGroundControl, Mission Planner, AMD Vivado